

FedICL-SQL: A Novel Federated Large-Small Language Models Framework with In-Context Learning for Natural Language to SQL

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Abstract. A concise (150–250 words) summary.

- Background
 - Importance of Text-to-SQL (Natural Language to SQL) systems
 - Challenges of distributed enterprise databases
- Problem
 - Data privacy constraints
 - Schema heterogeneity
 - High computational costs of LLMs
- Proposed Method
 - Federated LLM-SLM architecture
 - Schema-aware In-Context Learning
 - Federated knowledge distillation
- Results
- Practical deployment potential for enterprise Text-to-SQL systems

1. INTRODUCTION

- Natural Language to SQL
- Challenges
- Motivation
 - Federated Learning: Collaborative learning without data sharing.
 - Large Language Models: Strong reasoning and SQL synthesis.
 - Small Language Models: Efficient deployment.
 - In-Context Learning: Rapid adaptation to unseen schemas.
- Research Questions
 - RQ1 (Federated Learning effectiveness): Can FedICL-SQL improve Text-to-SQL performance in federated environments?
 - RQ2 (In-Context Learning effectiveness): Does schema-aware ICL improve cross-domain generalization?
 - RQ3 (Large-to-Small Language Model knowledge transfer and efficiency): Can LLM-to-SLM knowledge transfer achieve better efficiency-performance trade-offs?
- Research Objectives

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- To develop a novel privacy-preserving Text-to-SQL framework, namely FedICL-SQL, that integrates Federated Learning, Large-Small Language Model collaboration, and In-Context Learning to improve SQL generation performance, schema generalization, and deployment efficiency across distributed database environments.
- Main Contributions
 - Novel Federated LLM-SLM framework for Text-to-SQL.
 - Schema-aware In-Context Learning mechanism.
 - Federated SQL knowledge distillation strategy.
 - Privacy-preserving collaborative optimization.
 - Extensive evaluation on benchmark datasets.

2. RELATED WORK

2.1 Natural Language to SQL

2.2 Large Language Models for Text-to-SQL

2.3 Small Language Models

2.4 Federated Learning

2.5 In-Context Learning

2.6 Research Gap Analysis

3. PROPOSED APPROACH

3.1 Problem Formulation

3.2 Framework Overview

3.3 Global LLM Teacher

3.4 Local SLM Student

3.5 Teacher–Student Collaboration

3.6 Schema-Aware In-Context Learning

3.7 Federated SQL Knowledge Distillation

3.8 Federated Optimization

4. EXPERIMENTS

4.1 Experimental Setup

- Datasets:

- Spider
- Spider Realistic

- Teacher model: e.g., Qwen2.5-72B-Instruct

- Students models: e.g., Phi-3 Mini, Gemma 2B, TinyLlama,...
- Main baselines:
 - LLM-only models
 - SLM-only models
 - Centralized ICL systems
 - Without ICL
- Evaluation Metrics:
 - Text-to-SQL Metrics
 - Exact Match (EM)
 - Execution Accuracy (EX)
 - Federated Learning Metrics:
 - Communication Cost,
 - Convergence Rate
 - Efficiency Metrics:
 - Inference Latency,
 - GPU Memory Usage,
 - FLOPs
- Implementation details

4.2 Results & Discussion

- Overall Performance
- Impact of In-Context Learning
- Impact of LLM Teacher
- Impact of Federated Learning
- Communication Efficiency
- Privacy-Utility Trade-off
- Ablation Studies
 - Without ICL
 - Without federated learning
 - Without Teacher LLM
 - Different Retrieval Sizes
 - Different SLM Architectures

5. CONCLUSION

- Summary of FedICL-SQL
- Main findings
- Practical implications
- Future research directions
 - Federated Retrieval-Augmented Generation
 - Multi-database reasoning
 - Agentic SQL generation
 - Continual federated learning
 - Real-world enterprise deployment
